- Preparation of stock and working solutions in Laboratory
- Basic biochemical methods such as iodine test for polysaccharides
- Fermentation of sugars by Baker's yeast
- Isolation of amylose and amylopectin from starch
- Extraction of glycogen from liver
- Acid and enzymatic hydrolysis of glycogen
- Extraction and estimation of lipids from plant tissue/seed and lipid separation from different tissues
- Thin Layer Chromatography
- Macroscopic analysis of Urine
- Estimation of glucose levels in urine
- Fehlings Test for glucose estimation

Experiment No. 2

Preparation of stock and working solutions in Laboratory

Apparatus Required

Weighing balance, volumetric flask, Cylinder, beaker, stirrer, pH meter or litmus paper

Chemicals required

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Sodium chloride, Hydrochloric acid, sulfuric acid

Theory

Solution:

"A solution is a homogenous mixture of atoms, ions or molecules of two or more substances."

A homogenous mixture is that which has uniform position throughout its body.

Solvent and Solute

The solvent is the component of a solution that is visualized as dissolving another component called absolute. Usually the component present in the larger quantity is called the solvent, and the component present in the smaller quantity is called the solute.

Types of solution

Unsaturated solution

A solution that is capable of dissolving more solute at a given temperature than it already contains, is known as unsaturated solution.

Saturated solution

A saturated solution is the solution which can dissolve no more amount of the solute, at a given temperature.

Super saturated solution

A solution that contains more dissolved solute than a saturated solution is called super saturated solution.

Concentration and its units

Concentration means the relative amounts of the components of a solution. It tells the ratio of the quantity of one component to the quantity of the other or to the total quantity of solution. It has many units. Some common units are discussed below.

Mass Percentage

The ratio of the mass of the solute to the mass of the solution multiplied by 100 is called mass percentage.

$$\text{Mass Percentage of a solute} = \frac{\text{Mass of the solute} \times 100}{\text{Mass of solution}}$$

For liquid-liquid solutions, it is sometimes more convenient to express the concentration in the units of percentage by volume:

$$\text{Volume Percentage of a liquid} = \frac{\text{Volume of the liquid} \times 100}{\text{Total volume}}$$

Parts per Million (ppm)

This is used to express very dilute concentrations of a substance. One ppm is equal to 1mg of solute dissolved per litre of solvent or 1mg of solute dissolved per kg of solvent.

Molarity (M)

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Molarity or the molar concentration is the number of moles of solute dissolved per dm^3 of solution. ($1\ dm^3$ is equal to 1Liter)

$$\text{Molarity} = \frac{\text{Number of moles of solute}}{\text{Volume of solution in liters}}$$

Molality (m)

Molality is the number of moles of solute dissolved per kilogram of solvent.

Normality (N)

Normality is the number of equivalents dissolved per liter of solution.

$$\text{Normality} = \frac{\text{No. of equivalents}}{\text{Volume of solution in Litres}}$$

pH

The pH is defined as follows:

"The pH of a solution is the negative logarithm to the base 10 of the hydrogen ion (or Hydronium ion) concentration".

$$pH = -\log_{10}[H^+]$$

$$pH = -\log_{10}[H3O^+]$$

The pH of a neutral solution is 7, the pH of acids is less than 7 while that of bases is higher than 7.

Procedure

i. Preparation of 5% NaCl solution

1. Weigh 5grams of sodium chloride.
2. Dissolve the sodium chloride (NaCl) in 25 ml of distilled water in a beaker with the help of stirrer.
3. Transfer the contents into a volumetric flask. Add some more water (10-20ml) to the beaker and rinse the walls of beaker and add that to the same flask.
4. Raise the volume to 100 ml in the volumetric flask to prepare 5% NaCl solution.
5. Check the pH of the solution with pH meter.

Preparation of 5% hydrochloric acid solution (50ml) using 37% HCL.

Take 25 ml of distilled water in a volumetric flask.

Calculate the volume of HCL required by using the formula:

$$C1V1=C2V2$$

$$37xV=5x50$$

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V1=250/37

=6.7 ml

Add 6.7ml of HCl to the volumetric flask slowly and raise the volume to 50ml with distilled water to prepare 50 ml of 5%HCl solution.

Record the pH of the Solution using pH meter or litmus paper.

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Experiment No. 3

Basic biochemical methods such as iodine test for polysaccharides

Theory:

The iodine test is mainly performed to check the presence of carbohydrates among which starch and sugar are the main carbohydrates found in our food products.

Principles:

To produce the triiodide anion (I_3^-), elemental iodine is dissolved in an aqueous solution of potassium iodide. The resulting complex with starch produces an intense "blue-black" colour. The intensity of the colour decreases with increasing temperature and with the presence of water-miscible organic solvents such as ethanol. The test cannot be performed at very low pH due to the hydrolysis of the starch under these conditions. It is now thought that the iodine-iodide mixture combines with the starch to form an infinite polyiodide homopolymer. This was rationalized through single crystal x-ray crystallography and comparative Raman spectroscopy.

Material Required:

Knife, Spatula, Porcelain tile, Iodine solution and Food sample like potato or any other vegetable or fruit.

Procedure:

- Take a fresh potato which is washed, cleaned and dried. Peel off the skin of the potato as they are impermeable.
- Cut the potato into small cubes
- With the help of clean and dried spatula, place the potato sample on the clean and dried porcelain tile.
- Add 2 to 3 drops of dilute iodine solution on the potato samples.
- Keep the slides undisturbed and observe the change.

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Observations:

There will be a change in color. A blue black color develops on the slice or cubes of the potato samples.

Results:

According to the observation the food sample or the potato slice turned to blue-black on adding the iodine solution. This proves the presence of starch in the given sample.

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Experiment No. 04

Fermentation of sugars by Baker's yeast

Theory:

Yeasts are tiny, microscopic organisms—or microorganisms—that are actually a type of fungus. This means that they are more closely related to a mushroom than to plants and animals or bacteria (the latter of which are also microorganisms). These little critters might sound strange and different, but people have been using them for thousands of years to make bread rise. They turn this food into energy and release carbon dioxide gas as a result. This process is known as fermentation. When yeasts eat sugar and turn it into energy, they also produce carbon dioxide. This process is known as fermentation. In this activity, the balloons on the bottles should have captured carbon dioxide produced by the yeasts during fermentation.

Apparatus Required:

Beaker 500ml, 4 flasks of 250 ml, burner, tripod stand.

Reagents Required:

Yeast powder, water, Sugar.

Procedure:

- Take 5 flask and mark them A, B, C, D and E.
- Take 2.5g of yeast and put it in flask B, C and D.
- After that add 5g of sucrose in flask A, C and E.
- Add 10 gram of sucrose in flask D.
- Now add 100ml of distilled water to all flasks.
- Now boil only flask C for 5 minutes.
- Now close the mouth of all flasks with balloons.

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- Leave the whole assembly for 24 hours.

Results:

Balloons of flask A,B and C doesn't blown but the balloons of flask D and E will blow. This shows that fermentation occur in flask D and E during which CO2 evolved and become reason of blowing of balloons.

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Experiment No. 05

Isolation of amylose and amylopectin from starch

Theory:

Amylose is an un-branched chain polymer of D-glucose units. Amylopectin is a branched chain polymer of D-glucose units. It gives a dark blue/black color when iodine solution is added. It gives a reddish brown color when iodine solution is added.

Material Required:

Sweet potato, Knife, mortar & pastel, blender, blue capped tubes, 20mM sodium phosphate buffer at pH 7 and vortex

Procedure:

- Peel off sweet potatoes and note the weight.
- Beet sweet potato with the help of mortar and pastel
- Now blend sweet potato in blende by adding 40ml of cold sodium phosphate buffer in it.
- Blend until it converts into slurry.
- Transfer the slurry in blue capped tubes and mix with the help of vortex.
- Filter the extract with the help of cotton cloth or syringe filter.
- Centrifuge the filtrate at 12000rpm for 5min at 4 degree centigrade.
- After centrifugation discard the pallet and store the supernatant at 4 degree centigrade.

Results:

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The supernatant after centrifugation is consist of amylose and can be crystalized with the help of spray dryer in order to convert into powder form.

Experiment No. 06

Extraction of glycogen from liver

Theory:

Glycogen is a multibranched polysaccharide of glucose that serves as a form of energy storage in animals, fungi, and bacteria. The polysaccharide structure represents the main storage form of glucose in the body. Glycogen functions as one of two forms of energy reserves, glycogen being for short-term and the other form being triglyceride stores in adipose tissue (i.e., body fat) for long-term storage. In humans, glycogen is made and stored primarily in the cells of the liver and skeletal muscle. In the liver, glycogen can make up 5–6% of the organ's fresh weight, and the liver of an adult weighing 1.5 kg can store roughly 100–120 grams of glycogen. In skeletal muscle, glycogen is found in a low concentration (1–2% of the muscle mass) and the skeletal muscle of an adult weighing 70 kg stores roughly 400 grams of glycogen.

Material Required:

TCA, Centrifuge, glass cylinder, centrifuge tubes, 95% ethanol, water bath, Nacl

Procedure:

- Weight the liver sample and record the weight.
- Cut liver into small pieces and grind with about 0.5g of cold cold sand 10% TCA (1ml per g tissue).
- Centrifuge homogenate at 3000rpm for 5 minutes
- Pour off supernatant into ml graduated cylinder.
- Rinse out mortar with 5% TCA (using same volume as for 10% TCA already used)

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- Add this rinsing fluid to the centrifuge tubes containing residue from first centrifugation
- Re centrifuge for another 5 min at 3000rpm
- Discard pallet. Add supernatant already collected.
- Record the total volume; add the double of the supernatant volume of 95% ethanol, slowly with stirring to supernatant
- Allow to stand while precipitate settles, if not add a little Nacl and warm cylinder in water bath at 37 degree centigrade.
- Centrifuge supernatant at 3000rpm for 3 min. Discard supernatant.
- Now add 3ml diethyl ether, stir up pellet, recentrifuge and discard supernatant. This final pellet contains glycogen from liver.
- Air- dry the glycogen in the tube and weight it.

Results:

Weight the centrifuge tube that contains the glycogen = gm

Weight the empty centrifuge tubes used in experiment = gm

So Glycogen content (g) =

Centrifuge tubes contains pallet – empty centrifuge tube

Record total glycogen yield in 100g liver.

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Experiment No. 07

Extraction and estimation of lipids from plant tissue/seed and lipid separation from different tissues

Introduction:

Lipids play an important role in various cellular and physiological functions. Biochemical analysis of lipids requires their isolation. Depending on the type of the sample, extraction protocols vary considering the tissue structure, texture and lipid contents. The high sensitivity of some analytical methods employed to measure lipids requires the use of very pure solvents and clean glassware. Furthermore, all lipids must be protected against degradation through oxidation by solvent, oxygen, and enzymes in combination with temperature and light. Lipids can be broadly classified as polar and non polar lipids based on the head group present.

(Folch Extraction)

Requirements:

(1) Chloroform: Methanol (2:1, v/v), (2) 0.9% NaCl in water, (3) Homogeniser, (4) Lyophiliser

Procedure:

- The tissue is homogenised with Chloroform: Methanol (2:1, v/v) to a final volume 20 times the volume of the tissue sample (1 g in 20 ml of solvent mixture).
- Agitate the mixture for 15 min in an orbital shaker at RT.
- Centrifuge at 14,000 rpm for 10 min to get the clarified supernatant.
- Measure the volume of the supernatant and transfer to a fresh tube.
- Break phase by adding 0.2 volumes of 0.9% NaCl.
- Vortex the tube hard for 1 min.
- Centrifuge at a low speed of 2000 rpm to separate the two phases.
- Transfer the upper aqueous phase for analysis of ganglio- sides and polar molecules.
- Collect the lower organic phase containing lipids in a fresh tube.
- Evaporate in a lyophilizer or under a nitrogen stream and store below −80◦C.

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Experiment No. 08

Thin Layer Chromatography

Thin layer chromatography (TLC) consists of a thin layer adsorbent such as silica gel, alumina or cellulose on a flat carrier like a glass plate, a thick aluminum foil, or a plastic sheet. This layer of adsorbent acts as a stationary phase. This method is widely used in lipid analysis or is the standard method in organic chemistry for qualitative analysis of organic reactions. The adsorbent like silica has hydroxyl groups which act as interacting groups. The sample partitions between the mobile and the stationary phase. Individual components in the sample will interact with the stationary phase based on charge, solubility and adsorption. The components could then be visualised by iodine vapors or fluorescent dyes. The retention factor or the Rf value is a characteristic of the substance. This is a constant for a particular substance for that specific solvent and plate system. The Rf value is the ratio of the distance moved by the compound to that moved by the solvent.

Requirements:

(1) TLC plate
(2) Mobile phase solvent
(3) Glass jar

Procedure:

1) Bake the TLC plate in an oven set at 100◦C for 1 hr before use.

(2) Pour the appropriate developing solvent into a glass jar at least one hr before use. This is to saturate the jar with the running solvent vapors.

(3) Mark the TLC plate with pencil.

(4) Spot the sample and the respective standards onto the plate.

(5) Dip the plate in the running solvent just below the sample load.

(6) Allow the solvent to run due to capillary action till it reaches nearly the end of the plate.

(7) Remove the plate from the jar and let it dry.

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(8) Stain the plate for visualization of the compound.

(9) Measure the distance of the solvent and the compound travelled to obtain the Rf values.

Results:

Calculate the R_f by below given formula.

R_f = Distance travelled by the compound/ Distance travelled by the solvent front

For example, if a compound travels 2.1 cm and the solvent front travels 2.8 cm, the Rf is 0.75:

R_f = 2.1/2.8 = 0.75

Experiment No. 9

Macroscopic analysis of Urine

Key recommendations for urine sample collection and handling:

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- Patients should be well at baseline. They should have no urinary tract infection, no acute febrile illness, no intense exercise within the previous 24 hours.
- Recommended urine collection is a fresh, first morning void. A minimum of 5ml should be collected.
- If a first morning void is not practicable, random spot samples are acceptable.
- Samples not able to be delivered to the laboratory within 8 hours, should be refrigerated.
- Analysis should be performed on the day of receipt but samples can be stored for up to 7 days at 2-8 °C if necessary.
- Cloudy or particulate samples should be centrifuged prior to analysis.
- Positive ACR results must be confirmed, ideally on a fresh, first morning void, by repeat measurement on 1-2 occasions within 3 months.
- Prolonged storage should be at -70 °C; samples should not be stored at -20 °C.

Apparatus Required:

Plastic cup with a lid for sample collection, test tubes, test tube holder, dropper, beaker, litmus paper

Macroscopic Examination

The physical appearance of a urine sample can often tell a great deal about a patient's condition. A change in color or clarity may indicate the presence of a disease and the need for additional testing.

COLOR

Color is usually some shade of yellow and often varies with the concentration of the sample. The most common color descriptions of normal urine are straw, light yellow, yellow and dark yellow. Amber colored urine is seen in patients with increased bilirubin levels and may indicate hepatitis.

CLARITY

Clarity is an indication of the transparency of a specimen. It is best to judge clarity by observing light through a recently mixed sample. Terms used to describe clarity include clear, hazy, cloudy and turbid. Freshly voided urine that is properly collected is normally clear or slightly hazy, while contaminated urine is more likely to be hazy. Fresh urine that is cloudy is often the result of a bacterial urinary tract infection due to white blood cells in the urine. Turbid urine contains salt crystals that precipitate out as the specimen cooled.

pH

The pH is a measure of the degree of acidity or alkalinity of the urine. A pH below 7 indicates acidic urine; pH above 7 indicates alkaline urine. Normal, freshly-voided urine may have a pH

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range of 5.5 - 8.0. The pH of urine may change with diet, medications, kidney disease, and metabolic diseases such as diabetes mellitus. Colors on the pH reagent pad usually range from yellow-orange for acidic pH to green-blue when pH is alkaline.

Experiment 10

Fehlings Test for glucose estimation

Apparatus Required:

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Plastic cup with a lid, glass test tubes, holders, droppers, bottles for sample collection, test tube stand

Reagents Required

Fehling's solution A and B

Sample: Urine

Procedure:

In a test tube, add 2 ml of the test solution and add equal volumes of Fehling A & Fehling B and place it in a boiling water bath for few minutes.. When the contents of the test tube comes to boiling, mix them together and observe any change in color or precipitate formation. The production of yellow 'or brownish-red precipitate of cuprous oxide indicates the presence of reducing sugars in the given sample.

READINGS:

http://www.healthline.com/health/glucose-test-urine

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